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IoT Proposal

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Project Title: Mute Heart (An IoT-Enabled Wearable Gesture Recognition Glove for Sinhala Sign Language Voice Translation)

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Introduction

Human communication forms the foundation of social interaction, emotional expression, and access to services. For individuals with hearing and speech impairments, communication primarily occurs through structured visual languages such as Sri Lankan Sign Language (SSL). While SSL provides an effective medium within the deaf community, a substantial communication gap persists between sign language users and the broader population who lack SSL literacy.

Despite increasing research in gesture recognition systems, many existing implementations rely on camera-based approaches. Vision-based systems are often sensitive to illumination variation, background complexity, and computational overhead. In contrast, wearable sensor- based systems offer improved robustness, mobility, and real-time responsiveness.

Within the growing field of Internet of Things (IoT) wearable technologies, microcontroller-based gesture recognition gloves present a promising alternative for assistive communication systems. Leveraging embedded motion sensing and joint-angle detection enables translation of human gestures into machine-interpretable signals.

This research introduces **Mute Heart**, a right-hand wearable IoT glove that converts selected Sinhala sign language gestures into audible speech output. The system integrates motion sensing, angular displacement detection, embedded processing, and wireless communication within a compact assistive device. The proposed solution prioritizes affordability, portability, and real-time performance while maintaining architectural simplicity appropriate for scalable future enhancements.

Abstract

Communication barriers between hearing-impaired individuals and the public remain a persistent societal challenge, particularly in contexts where Sri Lankan Sign Language (SSL) is not widely understood. Although technological solutions for sign language interpretation exist, many are either vision-based systems susceptible to environmental conditions or high-cost sensor systems that lack portability and affordability.

This research proposes "Mute **Heart**", an IoT-enabled wearable gesture recognition glove designed to translate selected Sinhala sign language gestures performed using the right hand into audible Sinhala speech output. The system integrates an ESP32 microcontroller for data acquisition and wireless communication, dual potentiometers per finger for precise joint-angle detection, and an MPU9050 inertial measurement unit for three-axis motion capture. Gesture recognition is performed through rule-based signal threshold mapping, and the identified word is transmitted to a DFPlayer Mini audio module to generate pre-recorded Sinhala voice output via an embedded speaker. To support future integration of on-device machine learning algorithms for gesture classification, the system architecture allows optional deployment of a Raspberry Pi as an edge-processing unit for enhanced computational capability.

The proposed system emphasizes portability, cost-effectiveness, and real-time response, targeting a limited vocabulary of five Sinhala words selected from an initial pool of ten gestures. The device is powered using dual lithium-ion cells and constructed as a full wearable glove frame using either 3D-printed components or a mechanical support structure based on wooden sticks.

"Mute Heart" aims to provide an affordable assistive communication interface for the hearing-impaired community, contributing to inclusive human-technology interaction within the domain of IoT-based wearable systems.

Acknowledgement

We would like our deepest gratitude to everyone who helped to successfully complete this project. Firstly, we give our sincere appreciation to our lecturer, **Mr. Bathiya Seneviratne**, for guidance through our project and continuous support with his advice we completed our "**Mute Heart**" project. We are so grateful to our colleagues and our team members for their dedication to this project and the great communication between us and each member contributed to this project's success. Finally, we would like to show our acknowledgement to the National Institute of Business Management for providing necessary resources and the environment for this project.

Problem Definition

Although sign language serves as a structured communication system for the hearing-impaired community, widespread illiteracy in SSL among the general population results in frequent communication barriers. The limited availability of trained interpreters further intensifies this issue.

Existing technological solutions exhibit several limitations:

- Vision-based gesture recognition systems require controlled environments and high computational resources.
- Camera-dependent systems suffer performance degradation under variable lighting conditions.
- High-precision wearable sensor systems are often economically inaccessible.
- Many systems lack portability and real-time speech output capabilities.

Therefore, there exists a need for:

A **portable, cost-effective, real-time, sensor-based wearable system capable of translating selected Sinhala sign gestures into audible speech output without reliance on external vision infrastructure**.

Mute Heart is proposed to address this need by employing embedded IoT hardware and mechanical joint detection mechanisms integrated within a wearable glove structure.

Methodology

The proposed methodology follows a component-driven embedded system design approach consisting of sensing, processing, gesture classification, and speech generation stages.

3.1 System Architecture Overview

User Gesture

- Potentiometers Angle Detection
- Motion Data Capture (MPU9050)
- ESP32 Signal Processing
- Gesture Pattern Mapping
- Word Identification
- DFPlayer Mini Activation
- Speaker Output
- Raspberry Pi

3.2 Sensing Mechanism

Finger Joint Detection

Each of the five fingers of the right hand is instrumented with two rotary potentiometers, resulting in ten angular measurement channels. The potentiometers are mechanically coupled to finger joints using either 3D-printed structural holders or wooden stick-based lever mechanisms.

The dual-sensor configuration per finger improves angular resolution by capturing both proximal and intermediate joint movement, enabling more reliable gesture differentiation.

Motion Capture

An MPU9050 inertial measurement unit is integrated to capture three-axis acceleration and orientation data. This sensor enhances dynamic gesture recognition and compensates for hand orientation variations.

3.3 Processing Unit

The ESP32 serves as the central processing unit due to:

- Integrated Wi-Fi capability
- Sufficient analog input handling
- Low power consumption
- Real-time embedded processing capability

Analog signals from potentiometers are digitized through the ESP32 ADC channels. IMU data is collected via 12C communication.

Gesture classification is implemented using rule-based threshold mapping derived from calibration datasets for ten candidate Sinhala gestures, from which five will be selected for deployment.

Advanced Processing Extension (Optional Machine Learning Module)

To enhance the system's capability beyond rule-based gesture classification, the architecture allows integration of a Raspberry Pi 3 Model B+ as an auxiliary edge-processing unit.

While the ESP32 performs real-time signal acquisition and preliminary threshold-based classification, the Raspberry Pi can be deployed to execute computationally intensive machine learning algorithms for improved gesture recognition accuracy.

The Raspberry Pi enables:

- Execution of lightweight neural network models
- Implementation of supervised learning techniques
- Dataset logging and preprocessing
- Future deployment of adaptive learning models

In this hybrid architecture, the ESP32 functions as a low-level data acquisition controller, whereas the Raspberry Pi serves as a high-level intelligent processing module. Communication between the ESP32 and Raspberry Pi can be established via UART or Wi-Fi, ensuring modular system scalability.

This design ensures that the system remains functional in embedded mode while providing a pathway for future machine learning-based enhancement.

3.4 Voice Output Mechanism

Upon gesture recognition, the ESP32 transmits a digital command to the DFPlayer Mini module, which plays a pre-recorded Sinhala audio file corresponding to the detected word. Output is delivered via a compact embedded speaker.

3.5 Communication Capability

The ESP32 Wi-Fi module enables future IoT integration, such as:

- Cloud logging
- Mobile monitoring
- Remote data visualization

Although the primary output is audio-based, wireless capability ensures scalability.

3.6 Power Supply

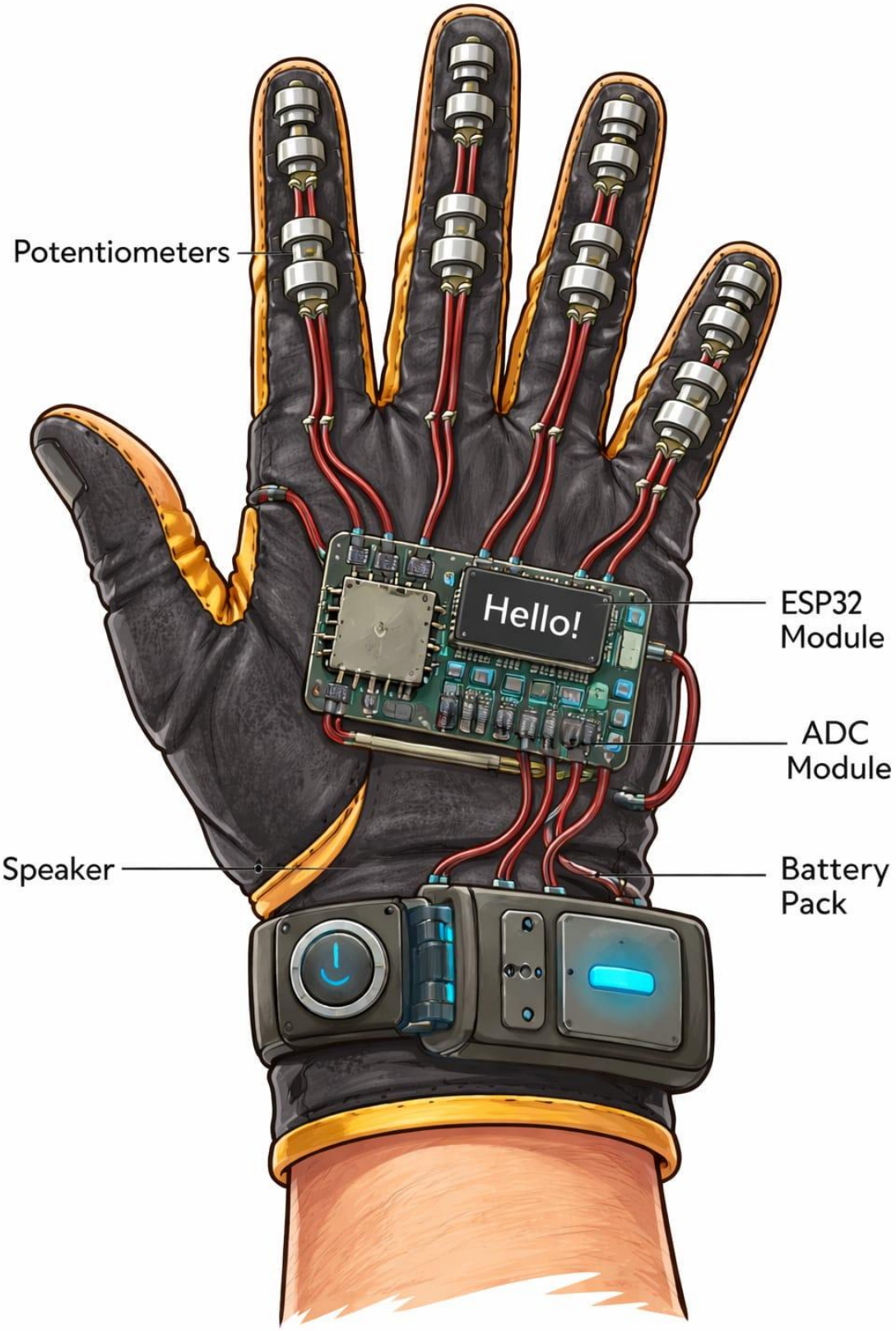
The system is powered using dual lithium-ion battery cells to ensure portability and sufficient operating duration.

3.7 Mechanical Structure

The glove is designed as a full wearable frame. Structural options include:

- 3D-printed PLA components for professional mechanical alignment
- Wooden mechanical linkage using rigid stick structures (fallback option)

The 3D-printed solution is preferred for precision and durability.


Potentiometers
Hello!
ESP32 Module
ADC Module
Speaker
Battery Pack

Components & Usage

Component	Functional Role
ESP32	Central processor, ADC reading, Wi-Fi communication
MPU9050	3-axis motion and orientation detection
Potentiometers (10 units)	Finger joint angle measurement
DFPlayer Mini	Audio playback control
Speaker	Audible word output
Li-ion Batteries	Portable power source
3D Printed Frame / Wooden Sticks	Sensor stabilization and structural integrity
ADC Expansion Module	Converts analog voltage signals from sensors into digital data and enables multiple sensor inputs to be processed accurately by the microcontroller
Raspberry Pi 3 Model B+	Acts as an advanced edge-processing unit to execute machine learning algorithms for gesture classification, perform dataset preprocessing, and enhance overall computational capability beyond embedded microcontroller limitations

References

These references helped us to support the design and implementation of the **Mute Heart** project.

[1] R. M. Rishan, S. Jayalal, and T. K. Wijayasiriwardhane, "Translation of Sri Lankan Sign Language to Sinhala Text: A Leap Motion Technology-Based Approach," *Proceedings of the 2nd International Conference on Advanced Research in Computing (ICARC)*, 2022.

[2] B. G. Lee and S. M. Lee, "Smart Wearable Hand Device for Sign Language Interpretation System with Sensors Fusion," *IEEE Sensors Journal*, vol. 18, no. 3, pp. 1224–1232, 2018. https://doi.org/10.1109/JSEN.2017.2779466

[3] M. A. Ahmed, B. B. Zaidan, A. A. Zaidan, et al., "Real-time Sign Language Framework Based on Wearable Device: Analysis of MSL DataGlove and Gesture Recognition," *Soft Computing*, Springer, 2021. https://doi.org/10.1007/s00500-021-05855-6

[4] R. Fatmi, S. Rashad, and R. Integlia, "Comparing ANN, SVM, and HMM-based Machine Learning Methods for American Sign Language Recognition Using Wearable Motion Sensors," *IEEE 9th Annual Computing and Communication Workshop and Conference (CCWC)*, 2019. https://doi.org/10.1109/CCWC.2019.8666491

[5] M. Oudah, A. Al-Naji, and J. Chahl, "Hand Gesture Recognition Based on Computer Vision: A Review of Techniques," *Journal of Imaging*, vol. 6, no. 8, 2020. https://doi.org/10.3390/jimaging6080073

[6] A. L. P. Madushanka, R. G. D. C. Senevirathne, L. M. H. Wijesekara, et al., "Framework for Sinhala Sign Language Recognition and Translation Using a Wearable Armband," *International Research Conference Publication*, 2020.

[7] D. Perera and S. Jayalal, "Sri Lankan Sign Language to Sinhala Text Using Convolutional Neural Network Combined with Scale Invariant Feature Transform (SIFT)," *International Journal of Computer Applications*, 2021.

[8] A. Weerasooriya and T. Ambegoda, "Sinhala Finger-Spelling Sign Language Recognition with Computer Vision," *IEEE International Conference on Moratuwa Engineering Research (MERCon)*, 2022. https://doi.org/10.1109/MERCon55799.2022.9906281

[9] H. G. Kortier, V. I. Sluiter, D. Roetenberg, and P. H. Veltink, "Assessment of Hand Kinematics Using Inertial and Magnetic Sensors," *Journal of NeuroEngineering and Rehabilitation*, vol. 11, 2014.

[10] Espressif Systems, "ESP32 Series Datasheet," Espressif Systems, 2023. https://www.espressif.com/en/products/socs/esp32

[11] P. Kumar, H. Gauba, P. P. Roy, and D. P. Dogra, "A Multimodal Framework for Sensor-Based Sign Language Recognition," *IEEE Transactions on Human-Machine Systems*, vol. 47, no. 4, pp. 576–588, 2017. https://doi.org/10.1109/THMS.2017.2688886

Conclusion

Mute Heart presents a structured IoT-based wearable assistive communication device designed to bridge the interaction gap between Sinhala sign language users and the general population. By integrating joint-angle detection, inertial motion sensing, embedded microcontroller processing, and real-time speech output, the system demonstrates a practical and scalable approach to low-cost sign language translation.

The proposed architecture prioritizes portability, affordability, and real-time responsiveness while maintaining modular design flexibility. Although limited to a five-word vocabulary in its initial implementation, the framework supports future expansion through enhanced gesture classification models, machine learning integration, and full alphabet coverage.

Mute Heart represents a meaningful step forward inclusive IoT-enabled human-machine interaction systems within assistive technology research.

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